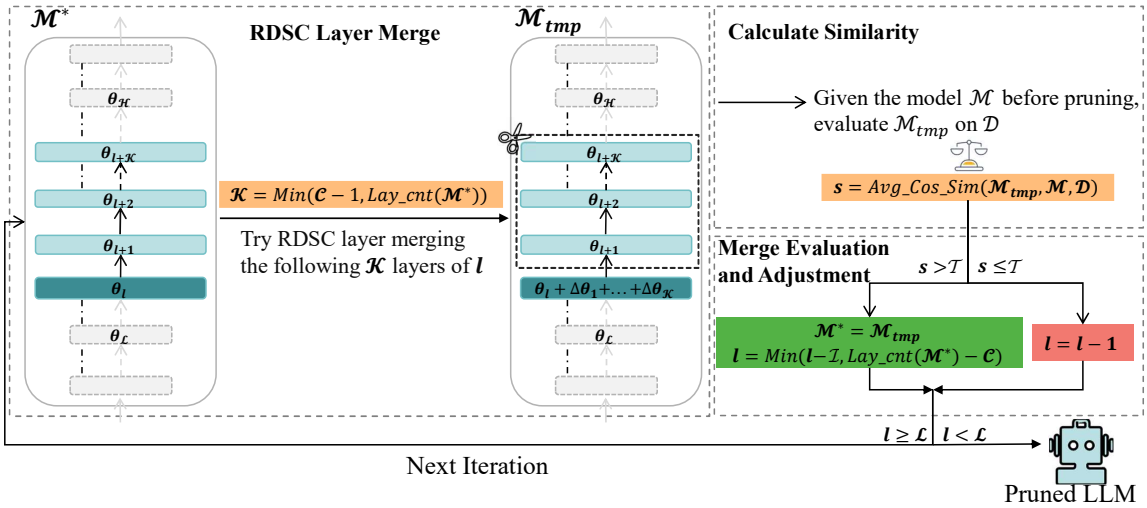

Figure 2: An illustration of Layer Collapse.

(2) Pruning (line 1-17)

We present an illustration of *Layer Collapse* in Figure 2. We begin by initializing the model $\mathcal{M}^*$ with the model $\mathcal{M}$ and set a layer pointer l to start from $\mathcal{H} - \mathcal{C}$. Then, the iterative process begins:

RDSC Layer Merge (line 4-5) During each iteration, our approach involves merging the $\mathcal{K}$ layers following layer l into layer l itself and then discarding the redundant $\mathcal{K}$ layers, where $\mathcal{K}$ is the minimum of $\mathcal{C} - 1$ and the total layer count of $\mathcal{M}^* - l$, implying merging either the subsequent $\mathcal{C} - 1$ layers or all layers following l, thus to prune the model $\mathcal{M}^*$, resulting in the interim model $\mathcal{M}_{tmp}$.

Calculate Similarity (line 6) We process each sentence in $\mathcal{D}$ using forward computations with $\mathcal{M}_{tmp}$ and $\mathcal{M}$ to derive their representations which are the output hidden-states of the last layer of the model. For every sentence, we then calculate the cosine similarity between these representations from both models, averaging these values to obtain the overall similarity score s.

Merge Evaluation and Adjustment (line 7-15) Then, we evaluate s against the threshold $\mathcal{T}$. Should s exceed $\mathcal{T}$, the current merge is considered successful. Then, $\mathcal{M}_{tmp}$ is updated to $\mathcal{M}^*$ for the next iteration, and the pointer l is adjusted downwards by $\mathcal{I}$ layers. Conversely, l is simply reduced by a single layer. It is important to highlight that the instances may occur where l falls below the total layer count of $\mathcal{M}^*$ after a series of successive merges. Consequently, it is required to reset l to the layer count in $\mathcal{M}^* - \mathcal{C}$, as illustrated in line 11.

We iterate through the above process until l is less than $\mathcal{L}$ and output the pruned LLM.

2.3 Complexity Analysis

The complexity of *LaCo* primarily depends on model inference. In the worst-case scenario, with $\mathcal{L}$ set to 0 and $\mathcal{H}$ to the total number of layers, if in each iteration the similarity s is less than $\mathcal{T}$, all layers will be traversed. Thus, the worst-case time complexity is $O(\mathcal{H} \times ||\mathcal{D}||)$. For example, for Llama2-13B with 40 layers and $||\mathcal{D}||$ consisting of 10 sentences, the maximum number of inference steps would be only 400 sentences, which can be completed within minutes on a single GPU.

3 Experiments

3.1 Models

To assess the effectiveness of the proposed *LaCo*, we conduct experiments on popular English LLMs, Llama2-7B and 13B (Touvron et al., 2023). Additionally, we test the effectiveness on bilingual LLMs, specifically Baichuan2-7B and 13B (Yang et al., 2023), which support both Chinese and English. We leverage the base versions of these LLMs. All these models are decoder-only models based on the transformer architecture.

3.2 Benchmarks

To comprehensively evaluate the pruned model's capabilities, we utilized the OpenCompass evaluation framework (Contributors, 2023). Specifically, following OpenCompass categorization, we conduct evaluations in five aspects: Reasoning, Language, Knowledge, Examination and Understanding. We select several benchmarks from each category. **Reasoning:** CMNLI (Xu et al., 2020), HellaSwag (HeSw) (Zellers et al., 2019), PIQA